

CS & IT ENGINEERING



Operating System

CPU Scheduling

Lecture -1

By- Vishvadeep Gothi sir



Recap of Previous Lecture



Topic

Process states

Topic

Process State Transition

Topic

Process Scheduling

Topic

Types of Schedulers

Topics to be Covered



Topic

Process Scheduling

Topic

Process Scheduling Algorithms

Topic

FCFS Scheduling

Topic

SJF Scheduling

#Q. Consider a system with n processes and m CPUs. Maximum and Minimum number of processes in each of the following states possible? $n > m$

1. Ready state	n	0
2. Running State	m	0
3. Blocked State	n	0
	max	min



Topic : CPU Scheduling



Function:

- Make a selection of process among ready process to run it next

Goal

- Minimize Wait time and Turn-around time
- Maximize CPU utilization (Throughput)
- Fairness





Topic : CPU Scheduling Types

Non-preemptive

Preemptive





Topic : Scheduling Times



- Arrival Time (AT): Time at which process arrives
- Burst/Service Time (BT): Amount of time process needs to run on CPU
- Completion Time (CT): Time at which process terminates
- Turn-Around Time (TAT): Amount of time process spends from arrival to completion
$$TAT = CT - AT$$
- Waiting Time (WT): Amount of time a process waits for CPU in ready state
$$WT = TAT - BT$$



Topic : Scheduling Times



- Response Time (RT): Amount of time from arrival to first time process gets CPU.





Topic : Scheduling Algorithms



1. FCFS
2. SJF
3. SRTF
4. LJF
5. LRTF
6. HRRN
7. Priority Based
8. Round Robin
9. Multilevel Queue Scheduling
10. Multilevel Feedback Queue Scheduling

Assumption :-

All processes do not have any I/O requirement.



Topic : FCFS (First Come First Serve)

the process which arrives first should be scheduled first.

Scheduling Criteria: smaller AT first | Tie breaker $\Rightarrow$ smaller process id first

Type of Algorithm: non-preemptive



Topic : FCFS (First Come First Serve)

Process	Arrival Time	^{CPU} Burst Time
P1	0	28
P2	0	4
P3	0	4

Gantt chart :- starts from 0





Topic : FCFS (First Come First Serve)

Process	Arrival Time	Burst Time	Completion Time	CT-AT	TAT-BT	RT
				Turnaround Time	Waiting Time	
P1	0	28	28	28	0	0
P2	0	4	32	32	28	28
P3	0	4	36	36	32	32

$$\text{avg TAT} = \frac{28 + 32 + 36}{3} = \frac{96}{3} = 32$$

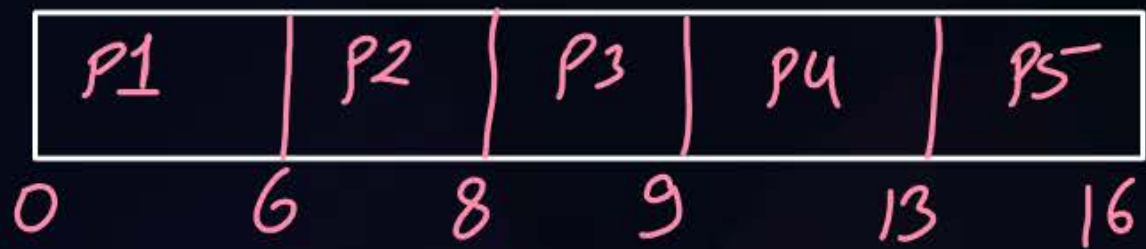
$$\text{avg WT} = \frac{0 + 28 + 32}{3} = \frac{60}{3} = 20$$

No. of context switches
b/w processes = 2
(Ignoring before first process
and after last process)



Topic : FCFS (First Come First Serve)

Process	Arrival Time	Burst Time	CT	TAT	WT	RT
P1	0	6	6	6	0	0
P2	1	2	8	7	5	5
P3	2	1	9	7	6	6
P4	3	4	13	10	6	6
P5	4	3	16	12	9	9



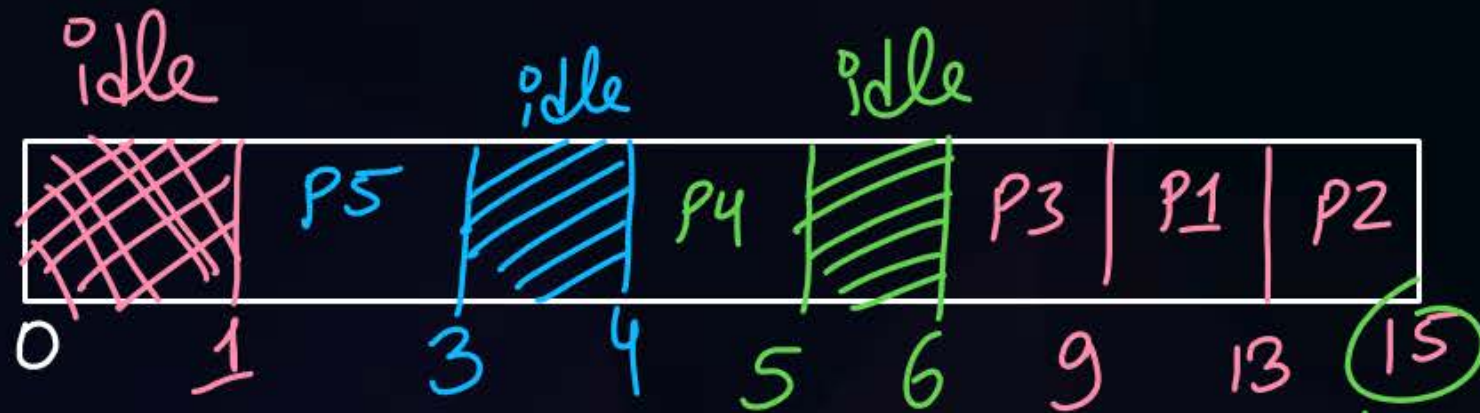
$$\text{avg TAT} = \frac{42}{5} = 8.4$$

$$\text{avg WT} = \frac{26}{5} = 5.2$$



Topic : FCFS (First Come First Serve)

Process	Arrival Time	Burst Time	CT	TAT	WT	RT
P1	7	4	13	6	2	2
P2	8	2	15	7	5	5
P3	6	3	9	3	0	0
P4	4	1	5	1	0	0
P5	1	2	3	2	0	0



Sum of all BT + Total idle time

$$\text{avg TAT} = \frac{19}{5} = 3.8$$

$$\text{avg WT} = \frac{7}{5} = 1.4$$

$$\% \text{ of time CPU remains idle} = \frac{3}{15} * 100\% = 20\%$$

$$\text{————||———— utilized} = \frac{12}{15} * 100\% = 80\%$$



Topic : Convoy Effect

If a big process is scheduled on CPU first, then all upcoming and following processes will have to wait for long-long time for execution.



it slows down system's performance.



Topic : FCFS (First Come First Serve)

Advantages:

1. Easy to implement
2. No complex logic
3. No starvation

Disadvantages:

1. No option of Preemption
2. Convoy effect makes the system slow

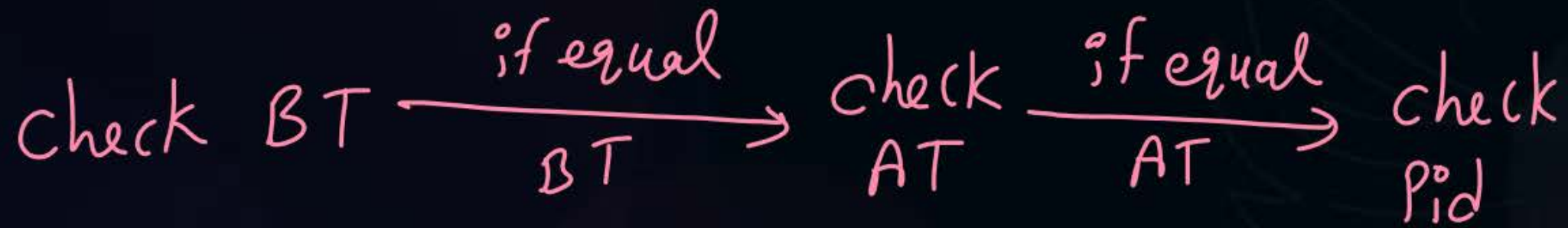


Topic : SJF (Shortest Job First)

run the process with smallest Burst time

Scheduling Criteria: smaller BT | Tie breaker $\Rightarrow$ FCFS

Type of Algorithm: Non-preemptive





Topic : SJF (Shortest Job First)

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time
P1	0	28	36	36	8
P2	0	4	4	4	0
P3	0	4	8	8	4



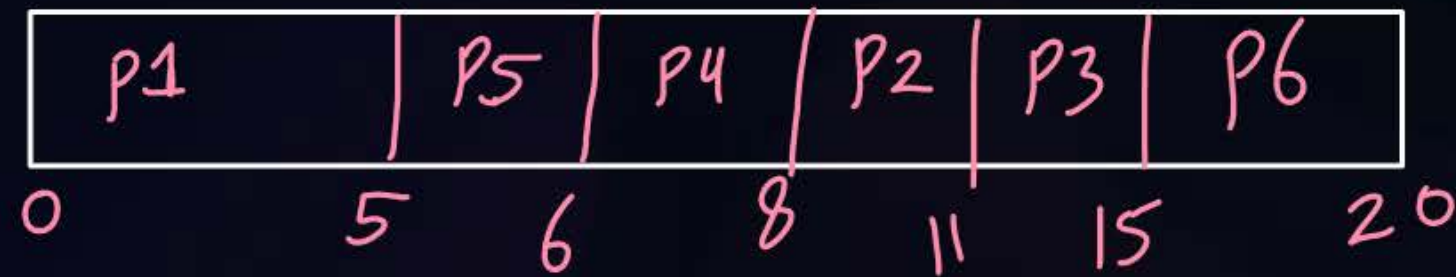
$$\text{avg TAT} = \frac{48}{3} = 16$$

$$\text{avg WT} = \frac{12}{3} = 4$$



Topic : SJF (Shortest Job First)

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time	RT
P1	0	5	5	5	0	0
P2	1	3	11	10	7	7
P3	2	4	15	13	9	9
P4	4	2	8	4	2	2
P5	5	1	6	1	0	0
P6	6	5	20	14	9	9



$$\text{avg TAT} = \frac{47}{6} = 7.8$$

$$\text{avg WT} = \frac{27}{6} = 4.5$$



Topic : SJF (Shortest Job First)

Advantages:

also min avg TAT

1. Minimum average waiting time among non-preemptive scheduling
2. Better throughput in continuous execution

↳ no. of processes executed in one time unit

Disadvantages:

1. No practical implementation because Burst time is not known in advance
2. No option of Preemption
3. Longer Processes may suffer from starvation

↳ indefinite waiting



2 mins Summary

Topic

Process Scheduling

Topic

Process Scheduling Algorithms

Topic

FCFS Scheduling

Topic

SJF Scheduling



Happy Learning

THANK - YOU

